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The principle of topological interlocking in extraterrestrial construction

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Abstract

Applications of a newly established principle of *topological interlocking* to different types of extraterrestrial construction are considered. Topological interlocking arises when elements of special shapes (usually convex or nearly convex, such that no stress concentration develops) are arranged in such a way that neither of them can be removed from the assembly without disturbing the neighbouring elements. Two types of extraterrestrial structures are considered. The first type represents mortar free structures built from specially engineered interlocking bricks, called *osteomorphic bricks*. The self-adjusting property of these bricks permits erecting structures which tolerate low precision of production and assembly, thus making the proposed method suitable for in situ produced bricks and low cost assembling machinery. The structures of the second type are modular extraterrestrial bases or space ships organised in topologically interlocking assemblies. For an extraterrestrial settlement such an organisation permits easy assembly even if the modules are uploaded on uneven ground. A space ship can be assembled from independent smaller ships interlocked topologically thus becoming a flexible vehicle suitable for both long-distance journeys and simultaneous exploration of extraterrestrial objects clustered in a relative proximity of each other.
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1. Introduction

With the increasing awareness for the need of utilisation and colonisation of the outer space in order to further economic, scientific and social developments, preserve Earth's ecology or even to ensure the survival of the human race (eg., [1–3]) the time is ripe to develop methods and techniques of extraterrestrial construction.

Currently, a number of architectures for the extraterrestrial structures are envisaged which can be broadly categorised into two major classes: (1) structures built from basic elements such as bricks, possibly manufactured in situ and (2) modular structures based on Earth-manufactured units (including structures assembled in orbit, e.g., [4]). The building elements of the extraterrestrial structures of the first class should satisfy the specific requirements which include easy delivery or

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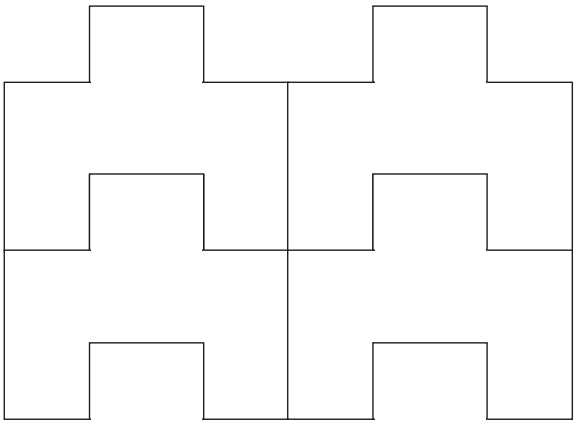

Fig. 1. Interlocking bricks proposed for lunar construction (adopted from [2]).

in situ manufacturing, easy assembly and multifunctionality. Attractive in this regard is the utilisation of interlocking blocks or bricks which can be assembled using special keys or connectors without the binder phase such for example as mortar. There are different types of interlocking blocks used in civil engineering (eg., [5–9]). A simple interlocking brick [10] (see also [2]) was proposed for lunar construction to be sintered in situ from the regolith (Fig. 1). An advantage of this technology, provided the current problems of sintering [2] are solved, is the abundance of the raw material and its high degree of radiation protection [11]. Interlocking is an important element of this concept since it removes the need for mortar and hence for two major ingredients—cement and water—the in situ production of which at industrial scale would require considerable capital investment. While cement can in principle be produced in situ, for example from regolith [12], supplying large quantities of water may pose a problem. In addition, as far as the construction on planets without an atmosphere is concerned, mortar will necessarily have to be cured in vacuum. This might be detrimental to the mortar quality due to creation of a porous structure inside the mortar layers caused for instance by intensive water evaporation [12] and concomitant reduction in density, strength and stiffness. (When concrete blocks are produced in situ, the effect of vacuum can in principle be avoided by precasting the blocks in a pressurised chamber. A similar approach to curing the mortar binder between the structural elements is obviously impractical.)

Traditional types of interlocking bricks, including the one proposed in [10] suffer however from a major drawback. As the bricks need to have special shapes to permit easy and preferably robotised assembly, they require quite precise manufacturing and maintenance of the gaps between the bricks, which may not be achievable in situ. In the lunar-type environment, the situation is further complicated by the need to handle the fine (finer than 20 μm) electrically charged fractions of lunar dust which levitate and electrostatically adhere to virtually everything [13,14] including the connecting surfaces of the bricks. Furthermore, under loading (both gravitational and thermal) the brick shape gives rise to considerable stress concentrations which are, in a long run, deleterious to the structural integrity of the lunar base.

Recently, a new type of interlocking, that we call topological interlocking, emerged that neither requires keys and connectors nor imposes high demands on the manufacturing precision. Based on this principle basic plate-like assemblies can be produced which form flexible mortarless structures with high fracture resistance and tolerance to missing blocks [15–17]. Interlocking of this type is achieved either by special arrangements of blocks of simple shapes (for instance platonic shapes [18]), or by specially designed curved surfaces [17]. An example of the first type of interlocking is given by plate-like assemblies of tetrahedra [19,20]. The second type of interlocking is exemplified by a set of so-called osteomorphic bricks, Fig. 2 [16,17,21] permitting construction of both plate-like and corner-like structures, as well as columns.

Possible applications of topologically interlocking blocks go beyond construction elements of structures of the first type. In extraterrestrial structures of the second type, the functional modules themselves can be shaped to ensure their interlocking, which opens up interesting design perspectives. This paper introduces the principle of topological interlocking and considers conceptually some applications in both planetary and space construction.

2. Principle of topological interlocking

Topological interlocking is achieved by a special choice of geometry and arrangement of the building elements such that neither element can be removed

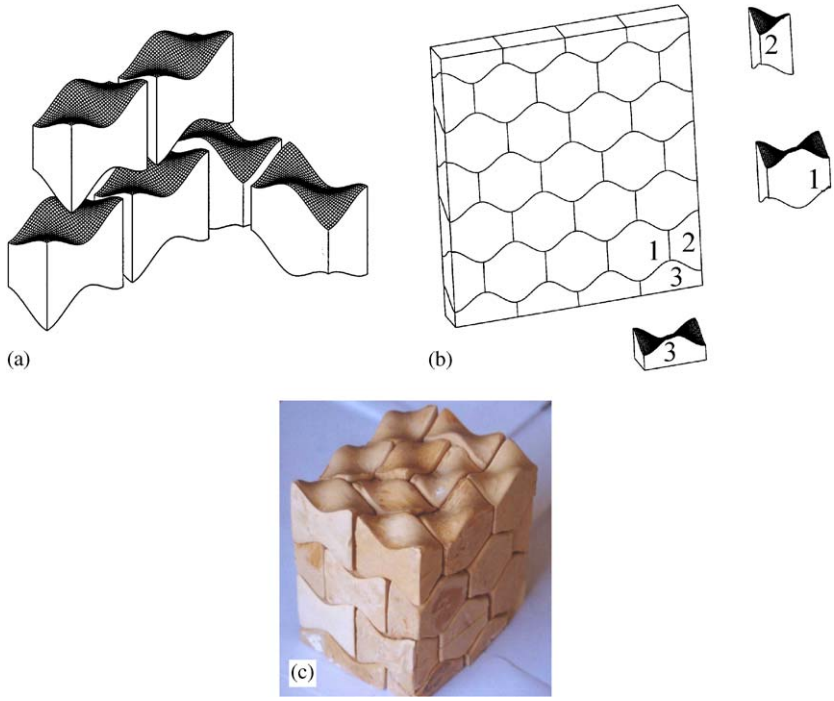

Fig. 2. Masonry structures assembled from osteomorphic bricks [17]: (a) principle of assembly of layer and corner structures, (b) layer of osteomorphic blocks (1) completed with half-blocks (2) and (3), (c) a column structure.

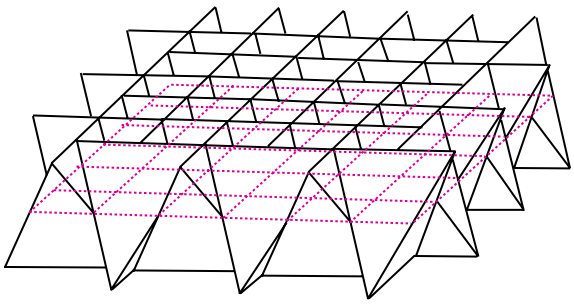

Fig. 3. A layer-like square-based structure of interlocked tetrahedra.

from the assembly without disturbing the neighbouring elements. In an assembly constrained at the periphery, each element will thus be locked.

Historically, the first interlocking assembly of this kind found was a plate-like assembly of tetrahedra [19,20], Fig. 3. The tetrahedra are put together in such a way that their middle sections form a square lattice covering a plane. It is seen on Fig. 3 that each tetrahedron has a pair of opposite faces inclined in one direction with the other pair inclined in another direction (the angle between the face and the vertical line is $\alpha = \sin^{-1}(\sqrt{3}/3) \approx 35.3°$). Therefore, any attempt to remove the element from the assembly results in pushing a couple of opposite neighbouring elements in-plane away from the element in question. If the movement of the neighbouring elements is prevented, for instance by applying an in-plane compression or peripheral constraint, the tetrahedral elements will be arrested within the structure.

Other examples are given in Figs. 4 and 5.

The main characteristic feature of the assemblies in the above examples is that the upward and downward (with respect to the assembly plane) movements of the elements are prevented by different faces. Interlocking can also be achieved through a single face of an element if the requirement of convexity is relaxed. This principle has been realised in osteomorphic blocks or bricks, Fig. 2 [16,17,21]. In the osteomorphic block each curved surface prevents its displacement along

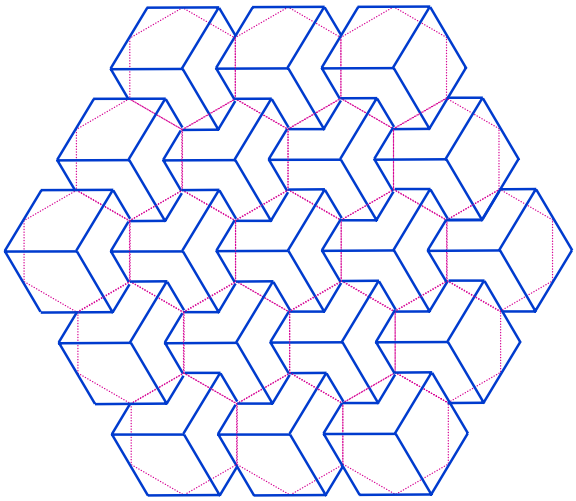

Fig. 4. A layer-like hexagon-based structure of interlocked cubes.

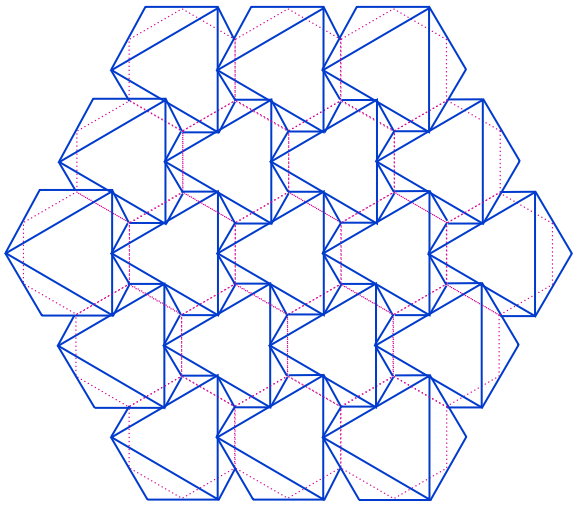

Fig. 5. A layer-like hexagon-based structure of interlocked octahedra.

the normal to the assembly in either direction. Moreover, these surfaces are engineered in such a way that both planar and corner structures are possible, making this block a convenient construction element. Also the shape of the interlocking surfaces makes the blocks self-adjusting such that any, even very inaccurate initial placement of a block, leads to its self-positioning into the correct place.

The above properties make the osteomorphic block a very attractive building element for extraterrestrial construction, as will be discussed in the following section.

3. Extraterrestrial construction based on osteomorphic bricks

3.1. Mortar free structures from osteomorphic bricks

As evident from the above, bricks made in the shape of the osteomorphic block allow erecting structures without mortar or any other binder. Another attractive property of these building blocks is that they are self-adjusting, such that the construction process does not require high precision positioning of the blocks, hence robotisation of the process can easily be achieved. Neither is high manufacturing precision required, so the construction can be based on in situ mass production of cheap building blocks (see Section 3.2 for discussion of the tolerance of the structure to variations in block dimensions).

These beneficial properties come at a price: the structure has to be either compressed or constrained as any other topologically interlocking structure. However, a clear advantage of the structures from osteomorphic bricks is that it only requires unidirectional constraint preventing the separation of curved faces of the bricks (in examples shown in Fig. 2, the constraint is in the vertical direction). This section discusses possible ways of providing this peripheral constraint as well as some construction and manufacturing methods.

3.1.1. Constraint by the natural features of planetary topography: foundations and in-built cavern structures

A natural peripheral constraint for plate-like interlocking structures can be provided by elements of planetary topography such as small craters or other pits. Fig. 6 shows an example of foundation built in a pit. The upper and lower gaps between the interlocking assembly and the pit walls should be either filled with an expanding grout (possibly mixed with regolith) or equipped with special jacks to provide the necessary constraining pressure (in Fig. 6 the grout is shown double-hatched). The other gaps (single line

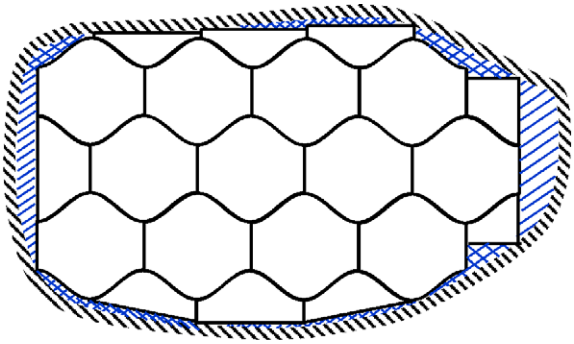

Fig. 6. Interlocking foundation built in a pit. The grey hatched area represents the pit wall; the upper and lower gaps between the walls and the structure (double-hatched) are filled with expanding grout. The other gaps may be left unfilled.

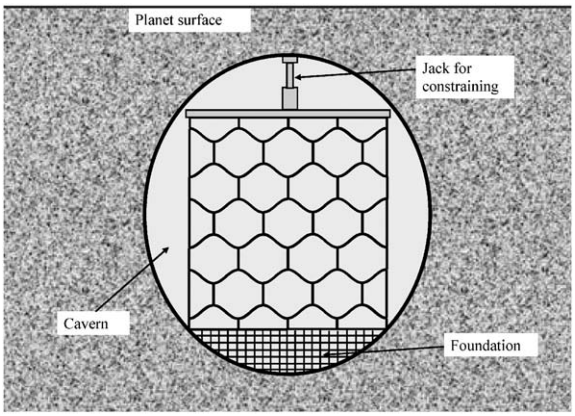

Fig. 7. Interlocking structure built in a cavern. The foundation can be built, e.g., from regolith.

filling in Fig. 6) can be left unfilled. In order to reduce the gaps, some bricks have been cut to shape fitting the pit geometry.

Another type of interlocking structures utilising the natural environment to provide the constraint are in-built structures in caverns or lava tubes. (For a general concept of utilisation of the lava tubes see [2].) Fig. 7 shows schematics of an in-built structure. In this case, only gaps between the structure top and the cavern roof should be either filled with expanding grout or equipped with the pressurising jacks (as shown in Fig. 7).

By the same token the interlocking structures could be fitted into larger artificial structures. However for a

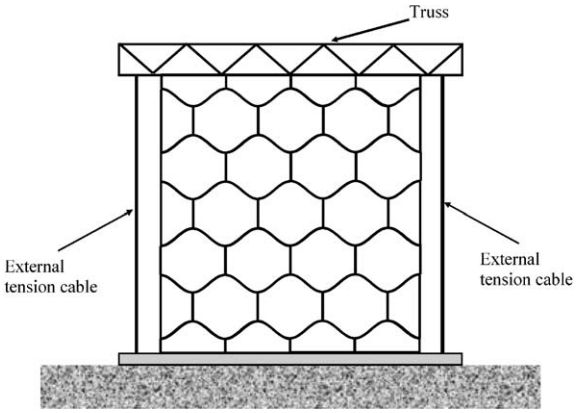

Fig. 8. An example of constraining by external frame.

standing alone structures other methods of providing the constraint are required. Some of them are discussed in the following subsection.

3.1.2. Artificial constraint

When natural constraint is not available, a specially built external frame can be used, as exemplified by Fig. 8. This design will, however, require the supply of pre-manufactured members, which in an early stage of extraterrestrial development not supported by an in situ industrial base will have to be transported from Earth. This may increase the development cost prohibitively.

A more viable solution can be a constraint by internal tension cables running through the bricks (at least two cables per brick), Fig. 9. After completion the structure the cables will have to be tensioned (so-called post-tensioning, e.g., [22]) in the way similar to tensioning tendons (high strength steel bars). The tensioning can be achieved by employing hydraulic jacks of different types. Tensioning can be conducted on a single cable at a time or on multiple cables simultaneously. After tensioning the cables have to be gripped. Generally there are three methods of gripping or anchoring (e.g., [23]). The first, most commonly used method is based on the wedge producing a frictional grip on the cable, Fig. 9. The second method uses the direct bearing from rivet or bolt heads formed at the end of the wires. The third method is based on looping the cables around the structure.

The post tensioning in every brick column is labour intensive, however it provides a low weight constraint

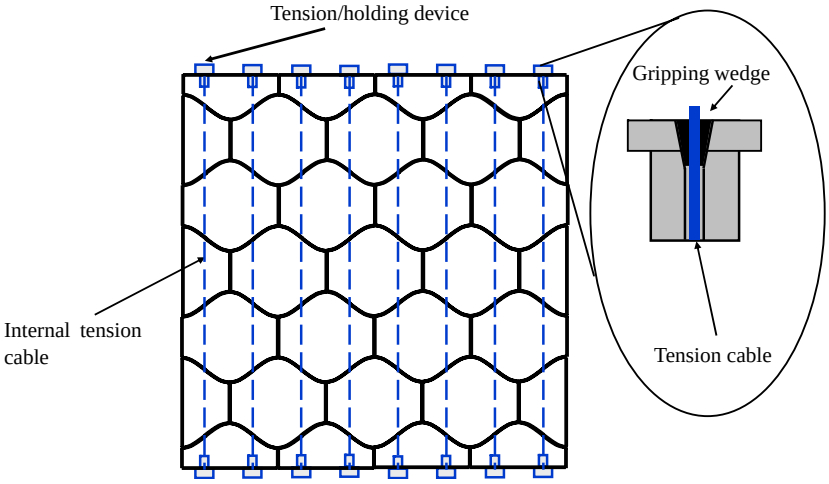

Fig. 9. Constraining by post-tensioning of internal cables.

(as compared to the external frame). Another advantage of this constraint method is its operational flexibility, since the number of cables and the density of their installation can vary within the same structure depending upon the constraint requirements.

3.1.3. Roofing

Roofing is typically achieved by construction elements other than the ones used for the main building. The advantage of the proposed osteomorphic bricks is that they can also be used to construct a roof, albeit only a planar one (special curved osteomorphic blocks have to be designed for other types of roofing, such as a dome type; alternatively, standardised planar 'cassettes' each consisting of a number of blocks can be used to approximate a curved surface of a dome). However, due to relatively low bending stiffness of assemblies of osteomorphic blocks [16], internal roof support is required in order to reduce the roof span and the associated inward deflection. It should be noted, however, that on planets with low gravity the achievable post-tensioning might be sufficient to minimise the deflections even in the case of relatively large roof spans.

3.1.4. In situ production of osteomorphic bricks

Although it cannot be ruled out that an efficient production technology based on cutting the bricks from bulk material, for instance mining them in quarries or cutting from large stones, can be viable, it seems that the main in situ production technology will be based on sintering of bricks from regolith dust.

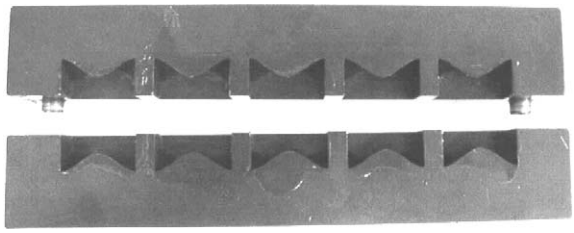

Fig. 10. Mould for simultaneous casting or sintering of 5 osteomorphic bricks [25]. The two separable halves of the mould are shown.

There appear to be two major sintering methods. One method—sintering in a mould by external or internal (microwave) heating—was already discussed (see [2]). There are still some problems with attaining the uniform temperature, so further work is required to develop the method to an industrial technology. Here we would only like to emphasise two points as far as the application of this method to producing osteomorphic bricks is concerned. Firstly, a special mould is required, cf. Fig. 10. The mould consists of two halves which need to be separated after sintering to enable the extraction of the brick. Secondly, it is important that the self-adjusting property of the bricks permits considerable tolerance to the quality of the surface such that after sintering no further machining is required.

Another possible sintering method is selective laser sintering used in rapid prototyping. In this method the bricks are sintered layer by layer with a precisely guided laser beam. For interlocking structures a clear advantage of this method is a possibility of simultaneous manufacturing of the whole layer of bricks already correctly positioned for interlocking [24]. An additional advantage is that no mould is required, which may reduce the transportation cost significantly, as otherwise a considerable number of moulds for industrial-scale brick manufacturing would have to be pre-fabricated and shipped to the planet where they are to be used. Presently, this method appears to be more time consuming than sintering in a mould, however, while in un-manned robotic missions time may not be an issue.

3.2. Sensitivity to manufacturing imperfections, planetary dust and weathering

When bricks are manufactured and used in the extraterrestrial environment, one has to face the possibility of surface imperfections inherent in the manufacturing process or stemming from weathering or planetary dust adherence. The fact that interlocking of the osteomorphic bricks is provided by the special curved faces rather than keys, pegs-and-hole connectors and the like makes the bricks self-adjusting. As a result, both the construction technology and the interlocking are insensitive to the imperfections of the contacting surfaces. From the structural point of view, the only important consequence of the imperfections is the inevitability of certain variations in the brick dimensions leading to unevenness of the upper edges of the constructed wall. In conventional brick and mortar structures it is the mortar layer that mitigates the brick size variations. In the interlocking structures the wall height variations will have to be dealt with by adjusting the tension in the cables. The necessary variations in the cable force can be estimated as follows. Let s be the standard deviation of the brick height, the average brick height being h. If the average wall (or structure) height is H, then the standard deviation of the wall height is $\Delta H = s(H/h)^{1/2}$ under a natural assumption that the size variations of different bricks are statistically independent. In order to compensate for this height variation one needs to produce a strain of $\Delta H/H$, which requires changing the tension cable force by

$$\Delta F \cong E\frac{sA}{\sqrt{hH}}, \tag{1}$$

where E is the effective Young's modulus of the structure and A is the horizontal cross-sectional area of the brick per cable (cf. Fig. 9). It is seen that the force variations required decrease with the wall height.

3.3. Radiation protection. Double skin walls

Bricks manufactured from the regolith can provide an efficient radiation protection owing to the high content of oxygen [11]. However, in mortar free structures the joints between the planar faces of the bricks are not filled and therefore can provide passage channels for radiation. In this case a double skin construction is necessary with the brick pattern of the outer skin being shifted with respect to that of the inner skin. If necessary, the cavity between the walls can be filled with bulk regolith to increase the thickness of the protective layer. Alternatively, a more advanced brick design can be used, with planar faces of the bricks being replaced with concave–convex ones, thus eliminating permeability with respect to radiation.

4. Protective cladding with osteomorphic tiles

The proposed osteomorphic bricks can be used for cladding pre-manufactured surfaces, for instance to shield a module or a spacecraft (see [26]). In this case the ability of the bricks to form corners may not be needed which removes the constraint on the brick thickness. For instance, it can be smaller than half of the brick length, as shown in Fig. 11. This type of applications may require cladding of *non-planar* surfaces, in which case non-planar osteomorphic blocks or tiles will be needed. The following procedure can be considered for generating the shapes of individual tiles to match the profile of the body of the spacecraft. Starting from a planar arrangement of the original planar blocks, a required area of the plane is mapped onto the spacecraft surface using a continuous function, followed by a three-dimensional continuation of the mapping function to transform the shape of each tile. A natural constraint in this case is that the tile

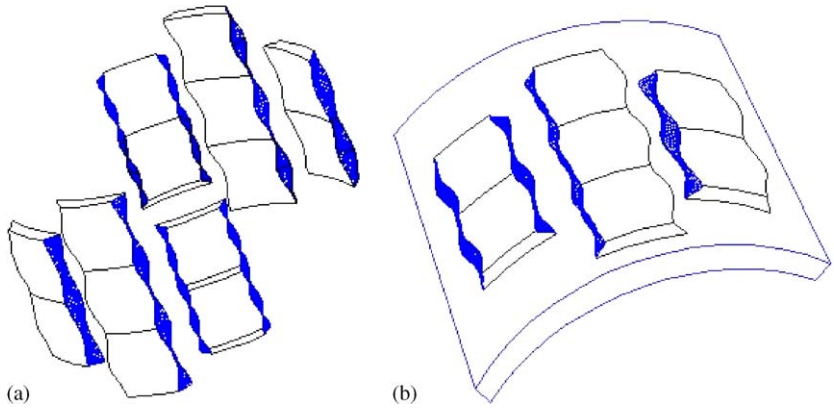

Fig. 11. Osteomorphic tiles for cladding of curved surfaces: (a) curved tiles and (b) tiles pre-positioned on a surface.

dimensions should be considerably smaller than the radii of curvature of the surface to be cladded.

Since cladding requires non-planar blocks with different curvatures and dimensions, the mass production method based on sintering in the mould may not be appropriate. Two other alternatives, the method of rapid prototyping or direct cutting from a bulk material will have to be used for the purpose.

Protective cladding with osteomorphic blocks has also a property of high fracture resistance as demonstrated by experiments presented in Fig. 12. Fig. 12a shows the interlocking assembly of rectangular shape of size $280 \times 260 \times 20\,\mathrm{mm}^3$ in a constraining frame. The thickness of the blocks (i.e. the thickness of the assembly) is 20 mm, other sizes are shown in Fig. 12a. The interlocking blocks were cast from cement paste (Portland cement with 38% water ratio) and cured for 21 days in a humidity room. The cup of 15 mm diameter in the centre of the assembly is a part of the indentor which delivers load to the structure. The structure rests on a flat surface which simulates the material to be protected.

Fig. 12b shows the fracture pattern produced by the indentation. It is seen that the fractures are contained within the loaded block and do not propagate into the neighbouring blocks. Fig. 12c shows the fracture pattern created in a reference solid plate of the same external size and the same material as the interlocking assembly. In this case the fractures do propagate over distances considerably larger than the block dimensions. In the reported series of tests the assembly failed at the average indentation load of 11.93 kN, while the solid plate failed at the average load of 9.9 kN, which is about 10% lower than that of the assembly.

5. Interlocking modular structures

5.1. Self-adjusting modules for an extraterrestrial base

Further possible interlocking configurations are based on regular polyhedra. It was demonstrated that all five platonic solids (tetrahedron, cube, octahedron, dodecahedron and icosahedron) can be arranged in interlocking layer-like structures [18]. Also, some ‘derivatives’ of the tetrahedron shape were proposed in [27]. Figs. 3–5 show the assemblies of elements that appear to be the most suitable ones for the purposes of extraterrestrial construction. Fig. 3 shows a square pattern of interlocked tetrahedral elements in the middle section of the assembly. Figs. 4 and 5 depict a hexagonal tiling in the middle section of the assemblies of interlocked cube- and octahedron-shaped elements. The main feature of these structures, viz. their non-smooth surface, makes them less suitable for construction purposes. However, at a larger scale, they can provide an interesting design for an extraterrestrial station. Indeed, if each element in Figs. 3–5 represents a pre-fabricated module of a station, then these structures can be thought of as a cluster type organisation of the station as per Reynolds’ classification (see [28]). The shapes shown on Figs. 3–5

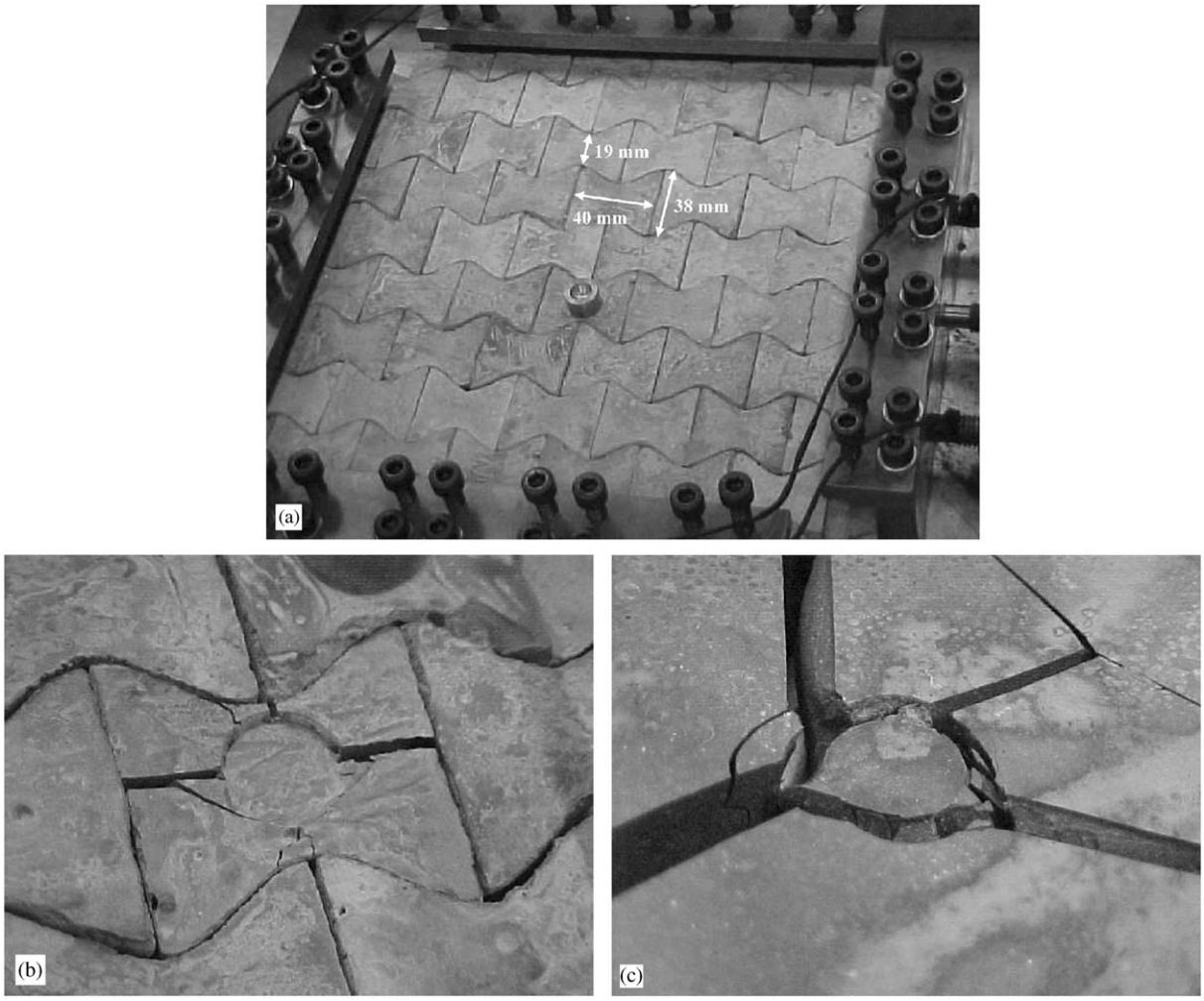

Fig. 12. Fracture resistance of assembly of osteomorphic blocks vis-à-vis that of a solid plate: (a) the osteomorphic assembly in a constraining frame; the cup in the middle is a part of the indenting device, (b) fracturing of the osteomorphic assembly is contained within the loaded block and (c) unconstrained fracturing of a solid reference plate.

are, of course, not the only possible ones. The osteomorphic shape could also be suitable for the purpose; however, manufacturing of modules of such a shape is more involved.

The advantage of using the interlocking organisation of modules is in the inherent self-adjusting property of topological interlocking. The same geometry that locks the elements within the structure ensures that if in a loose structure the elements are not properly aligned, application of lateral load will force them into a regular planar arrangement. This is illustrated by Fig. 13 where the interlocking is indicated by two parallel sections (in reality the sections that ensure interlocking do not have to be parallel—they are shown parallel for illustrative purposes only). Fig. 13a sketches a collection of modules after they have been uploaded on an uneven ground. Then a lateral force needs to be applied to assemble them into a cohesive structure, Fig. 13b. Thus, one can envisage an express method of station assembly on planets with low gravity which does not require special ground preparation.

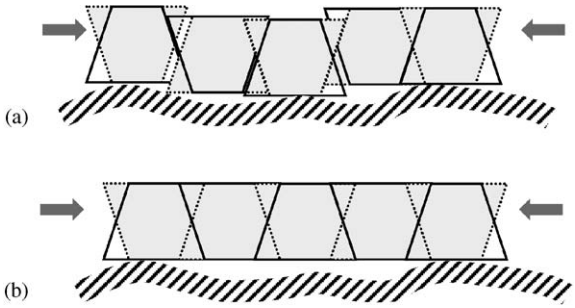

Fig. 13. Construction on uneven surface using the self-adjusting property of the interlocking modules. Here for illustrative purposes interlocking is indicated by two parallel sections: (a) initial positioning of the pre-fabricated modules uploaded on uneven ground; the arrows indicate the assembling force applied to the collection of modules and (b) the resultant assembly.

Suppose that the angle between a face of contact and the vertical line is α, while the friction angle for contacting faces is φ and cohesion can be neglected. Then, in order to lift a module of weight P, a lateral force given by

$$F = \frac{P}{n} \cot(\alpha - \varphi), \tag{2}$$

needs to be applied to the neighbouring modules, with $n = 2$ for square-based assemblies and $n = 3$ for hexagon-based assemblies. The total force necessary for assembling the structure will depend upon the structure configuration, the number of the modules to be lifted, their mass and the planet's gravity. Eq. (2) shows that there is a critical parameter in this method, viz. the friction angle. For the method to work, the friction angle must be smaller than the face inclination angle. (Already when $\varphi = \alpha$ the force needed to lift a module becomes infinite.) For the structures discussed here the inclination angles are $\alpha = 35.3°$ for tetrahedra and cubes and $\alpha = 19.5°$ for octahedra. Therefore, assemblies of tetrahedra or cubes are more preferable from this point of view.

5.2. Removing of modules from the structure

Another aspect of the topological interlocking is that it can give rise to assemblies with high tolerance to missing blocks. The aforementioned structures differ in this regard. The structures assembled from osteomorphic blocks (Fig. 2) as well as the square-tiling based structures (Fig. 3) are tolerant to missing blocks, i.e. the structural integrity of the assemblies are maintained even if isolated individual blocks are removed [16]. On the other hand, the hexagon-tiling based structures (Figs. 4 and 5) do not tolerate missing blocks and disintegrate once a single block is removed or fails. This imposes certain restrictions on the geometric shape to be chosen for interlocking assemblies of modules of a station. In the case when the overall integrity is the main issue, and loss of part of the modules is acceptable, either square-tiling based or osteomorphic blocks should be used. If however, easy replacement of the modules is important, then one should use the hexagon-tiling based structures, since in that case each module can be accessed by removing modules one by one via the shortest route leading to it, Fig. 14.

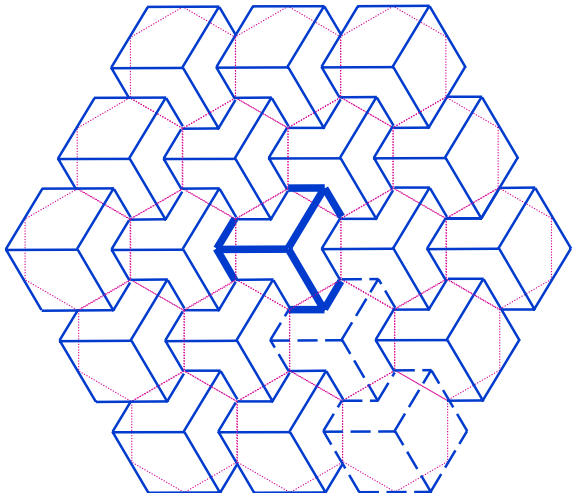

Fig. 14. A path to service a faulty module (highlighted): broken lines show the modules that should be removed to access the faulty module.

5.3. Spaceship based on interlocked modules

Another possible application of interlocking structures is the in-orbit assembly of a spacecraft from independent modules. In this case the thrust shall be used to keep the modules together by inclining the jets slightly such that they generate a small in-plane force, Fig. 15. The interlocking structures are quite flexible [20], which will provide additional shock absorption for central modules that can be used as living quarters.

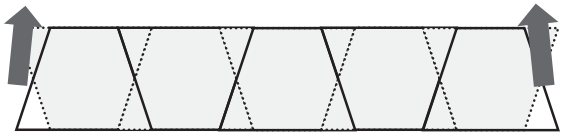

Fig. 15. Schematics of a spacecraft assembled from independent modules. The arrows indicate the direction of the thrust (upward in this illustration): a small horizontal component is needed for integrity of the spacecraft structure to be maintained.

This type of organisation of the spacecraft opens up yet another interesting possibility. Each module can be equipped with its own engine and thus turned into a small spacecraft in its own right. Then at the object of interest (e.g., a planet with satellites) the modular spacecraft can be disassembled into separate spacecraft-modules to explore different parts of the object. After finishing the exploration the modular spacecraft can be reassembled for further interplanetary journey. A particular design could for instance provide the landing capability to only some of the modules, while equipping the peripheral modules with special engines to provide the main thrust for distant journeys.

6. Conclusion

The principle of topological interlocking offers new possibilities for extraterrestrial construction:

1. Various mortar free and demountable structures can be built from osteomorphic bricks. The osteomorphic bricks are self-adjusting, which relaxes the accuracy requirements for both the production and the assembly of the bricks, making the method ideal for in situ production and robotisation. In addition, the interlocking structures from osteomorphic bricks possess high resistance to fracture propagation and are tolerant to missing bricks. A peripheral constraint necessary for stability of such structures can be provided either by utilising natural features of planetary topography or by artificial means, e.g. installation and post-tensioning of internal cables or erecting an external constraining frame.

2. A further possible application of interlocking structures is in protective cladding (e.g., for thermal or radiation protection) based on osteomorphic tiles made from a suitable material.

3. Modular structures from interlocked modules (for instance in the shape of tetrahedra, cubes or octahedra) can be advantageous because their self-adjusting property permits easy assembling by applying lateral force to a group of uploaded modules even if the ground they are resting on is uneven.

4. Spacecraft can be built as interlocking assemblies of a number of independent spacecraft as modules. This offers a highly flexible multipurpose vehicle suitable for both long-distance travel and simultaneous exploration of extraterrestrial bodies clustered in proximity of each other.

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